# Plan 3 Single-Level Recheck and the Planar Annular Bridge

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## 1 Outcome

The sound Plan 3 information is more useful than the old graph route, but it does not yet close a new proof.

- The Chebyshev good-level extraction, the deficit transfer from $\Omega$ to $E_{\hat{t}}$, and the final superlevel-to-domain Fraenkel transfer are sound quantitative steps.
- The recently proved small-amplitude radial-graph theorem is also sound: for star-shaped radial graphs in a thin annulus it gives

  $$E(D) - E(B) \geq c_n \|R - 1\|^2_{L^2(\mathbb{S}^{n-1})} \geq c'_n \mathcal{A}(D)^2$$

  with no slope, curvature, $C^1$, or perimeter assumption.
- The single-level Hessian/Weinberger "Chain B" currently recorded in `proof-step2.md` and `verify-chainB.md` is not sound. The exact false point is the unweighted Hessian domination

  $$\delta_T(D) \geq c_n |D|^{-1} \int_D \left| D^2 u_D + \frac{1}{n} I \right|^2 .$$

  This has the wrong derivative count and fails on smooth nearly spherical high-frequency perturbations.
- In dimension 2, the most promising non-Plan-1 route is therefore: good level $\Rightarrow$ centered small annular containment, followed by a new annular-set stability theorem modulo translations. Neither implication is currently proved in the repo. The first is only recorded as a reconstructed status note; the second is not present at all.

## 2 The precise overclaim: unweighted Hessian control

The claimed Chain B uses the following statement, labelled $(T)$ in `Plan 3/C-assembled-proof-steps/proof-s`

$$\delta_T(D) \gtrsim_n |D|^{-1} \int_D \left| D^2 u_D + \frac{1}{n} I \right|^2 . \tag{T}$$

The verification note `Plan 3/B-routeB-weinberger/verify-chainB.md` then treats this as a regularity-free Weinberger quantitative inequality. This is the point where compactness/regularity was not eliminated; a much stronger and false Hessian statement was substituted for the actual Saint–Venant deficit.

**Proposition 2.1** (High-frequency obstruction to $(T)$)**.** *There is no dimensional $c_n > 0$ such that $(T)$ holds for all smooth volume-normalized near-balls.*

*Proof.* Let $Y_k$ be an $L^2(\mathbb{S}^{n-1})$-normalized spherical harmonic of degree $k \geq 2$. Consider the smooth radial graph

$$D_{\varepsilon,k} = \{r < R_{\varepsilon,k}(\theta)\}, \qquad R_{\varepsilon,k} = 1 + \varepsilon Y_k + O_k(\varepsilon^2),$$

where the $O_k(\varepsilon^2)$ constant mode enforces volume and no first-mode correction is needed because $k \geq 2$. The expansion below is obtained by the elementary ball linearization, equivalently by the exact radial-graph identity in `H12_ASYMMETRY_DIRECT_PROOF.tex`; no BDV theorem is being used.

Set

$$H_k(r,\theta) = r^k Y_k(\theta).$$

The first variation of $u_D - u_B$ is $\varepsilon H_k/n$. The torsion deficit has the Steklov, half-derivative scale

$$\delta_T(D_{\varepsilon,k}) = \frac{\varepsilon^2}{2n^2}(k-1) + O_k(\varepsilon^3). \tag{2.1}$$

Indeed the harmonic energy contributes $\frac{1}{2}(\varepsilon^2/n^2)\int_{B_1}|\nabla H_k|^2 = \frac{\varepsilon^2}{2n^2}k$, while the volume correction in the moment term contributes $-\varepsilon^2/(2n^2)$.

By contrast, the unweighted Hessian defect has two more powers of frequency. Using

$$\Delta|\nabla H_k|^2 = 2|D^2 H_k|^2,$$

and

$$|\nabla H_k|^2\big|_{\partial B_1} = k^2 Y_k^2 + |\nabla_\tau Y_k|^2, \qquad \int_{\mathbb{S}^{n-1}} |\nabla_\tau Y_k|^2 = k(k+n-2),$$

we get

$$\int_{B_1} |D^2 H_k|^2 = (k-1)\big(k^2 + k(k+n-2)\big) = (k-1)k(2k+n-2). \tag{2.2}$$

Therefore

$$\int_{D_{\varepsilon,k}} \left|D^2 u_{D_{\varepsilon,k}} + \frac{1}{n}I\right|^2 = \frac{\varepsilon^2}{n^2}(k-1)k(2k+n-2) + O_k(\varepsilon^3). \tag{2.3}$$

Taking $\varepsilon \downarrow 0$ for fixed $k$, then $k \to \infty$, gives

$$\frac{\delta_T(D_{\varepsilon,k})}{\int_{D_{\varepsilon,k}} |D^2 u_{D_{\varepsilon,k}} + \frac{1}{n}I|^2} \longrightarrow \frac{1}{2k(2k+n-2)} \longrightarrow 0.$$

Thus no dimensional lower bound $(T)$ can be true. $\square$

The weighted Hessian variant is also false at the sharp scale, but less violently: $\int_{B_1} u_B |D^2 H_k|^2 = k(k-1)/n$, so it sees an $H^1$ boundary scale. The unweighted Chain B sees an even stronger scale and therefore cannot be the missing Plan 3 closure.

# 3 What remains sound

The following pieces survive this recheck.

1. **Step 1.** The good-level extraction gives a Sard-regular level $E_{\hat{t}}$ with small $D_H$-integrand and
$$|\Omega \setminus E_{\hat{t}}| \leq C_{n,\theta}\, \delta_T(\Omega).$$
The important patch is that the argument is made in the volume variable $s = m(t)$, where the layer size is exactly linear.

2. **Step 3.** The function $u_\Omega - \hat{t}$ is exactly the torsion function of $E_{\hat{t}}$. The profile computation gives

$$\delta_T(E_{\hat{t}}) \le \frac{1}{1 - |\Omega \setminus E_{\hat{t}}|/|\Omega|}\, \delta_T(\Omega),$$

hence $\delta_T(E_{\hat{t}}) \le 2\delta_T(\Omega)$ in the small-deficit regime.

3. **Step 5.** For $E_{\hat{t}} \subset \Omega$,

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\hat{t}}) + 2\frac{|\Omega \setminus E_{\hat{t}}|}{|\Omega|}.$$

After squaring, the transfer error is $O(\delta_T(\Omega)^2)$.

4. **Small-amplitude radial graphs.** The direct radial-graph proof closes the desired $A^2$ theorem in a thin annulus for genuine radial graphs. This is stronger than the old graph route in one direction: it does not need any slope or curvature bound. It still needs one radial crossing per direction.

# 4 The planar route that is still plausible

The $n = 2$ note `n2-graph-route-OBSTRUCTED.md` records the right positive target: near-Serrin data on a planar level should perhaps give only $L^\infty$/inner–outer radius control, not $C^1$, $H^1$, or curvature control. But this is a reconstructed verdict, not a written proof in the repo. With the new radial-graph theorem, such a weak conclusion would no longer be obviously too weak. The graph theorem only needs small amplitude, not small slope.

There are therefore two separate missing statements. First, one needs the annular entry lemma.

**Conjecture 4.1** (Planar annular entry from the good level)**.** *Let $E_{\hat{t}} \subset \mathbb{R}^2$ be the Sard-regular level obtained by the Plan 3 Step 1 extraction, with normalized area and $D_H(\hat{t}) \le \theta$. There exist a center $x_0$ and a modulus $\eta(\theta) \to 0$, with computable constants, such that*

$$B_{1-\eta(\theta)}(x_0) \subset E_{\hat{t}} \subset B_{1+\eta(\theta)}(x_0). \tag{AE}$$

This would be weaker than graph entry and is exactly why dimension 2 is still worth pursuing. But (AE) is not proved in the current notes.

There is one neutral-mode warning before formulating the second statement: the fixed-center inequality

$$E(D) - E(B_1) \ge c\,|D \Delta B_1|^2$$

is false under only $B_{1-\eta} \subset D \subset B_{1+\eta}$. A translated disk $D = B_1(a)$, $0 < |a| \le \eta$, satisfies the annular containment and has $E(D) = E(B_1)$, but $|D\Delta B_1| > 0$. The bridge must either impose a centering condition eliminating the first Fourier mode, or state the right side with the Fraenkel infimum over translated balls.

Second, even if (AE) were available with the correct center, annular containment is not the same as being a radial graph. A level set may have several crossings on one ray, holes/fjords inside the annulus, or same-ray cancellation between an inward missing piece and an outward excess piece. Thus the second missing statement is the following.

**Conjecture 4.2** (Planar annular bridge, centered form)**.** *There exist computable constants $\eta_0, c_0 > 0$ such that, if*

$$|D| = \pi, \qquad B_{1-\eta} \subset D \subset B_{1+\eta}, \qquad \int_D x\,dx = 0, \qquad 0 < \eta \le \eta_0,$$

*then*

$$E(D) - E(B_1) \geq c_0 \, |D \Delta B_1|^2. \tag{AB}$$

*Equivalently, in translation-invariant language, one may replace the right-hand side by* $c_0 \inf_a |D \Delta B_1(a)|^2$*, provided the annular entry lemma also supplies a compatible center.*

If both (AE) and (AB) are proved with compatible centers, the planar single-level route closes without BDV: Step 1 picks $E_{\hat{t}}$ with a small $D_H$-level; the planar annular-entry lemma gives

$$B_{1-\eta(\theta)} \subset E^{\sharp}_{\hat{t}} \subset B_{1+\eta(\theta)}, \qquad \eta(\theta) \lesssim \theta^{1/2};$$

choose the fixed $\theta$ so that $\eta(\theta) \leq \eta_0$; Step 3 gives $\delta_T(E_{\hat{t}}) \leq C\delta_T(\Omega)$; (AB) gives $\mathcal{A}(E_{\hat{t}})^2 \leq C\delta_T(E_{\hat{t}})$; Step 5 transfers this back to $\Omega$.

# 5 Attempted proof of the annular bridge

There is a clean identity for arbitrary planar annular sets which identifies the real issue. Let $D \subset \mathbb{R}^2$ be smooth, $|D| = \pi$, and

$$u_0(x) = \frac{1 - |x|^2}{4}, \qquad z = u_D - u_0.$$

Then $z$ is harmonic in $D$ and $z = -u_0$ on $\partial D$. Expanding the torsion energy around $u_0$ gives

$$E(D) - E(B_1) = \frac{1}{2} \int_D |\nabla z|^2 \, dx - \frac{1}{8} \left( \int_D |x|^2 \, dx - \int_{B_1} |x|^2 \, dx \right). \tag{4.1}$$

For $B_{1-\eta} \subset D \subset B_{1+\eta}$, the moment term is

$$M_2(D) := \int_{D \setminus B_1} (|x|^2 - 1) \, dx + \int_{B_1 \setminus D} (1 - |x|^2) \, dx \geq 0,$$

so (4.1) reads

$$E(D) - E(B_1) = \frac{1}{2} \int_D |\nabla z|^2 - \frac{1}{8} M_2(D). \tag{4.2}$$

The geometric side is also clean once the translation mode has been fixed. With

$$M_1(D) := \int_{D \Delta B_1} \big| |x| - 1 \big| \, dx,$$

the annular bounds imply $M_2(D) \simeq M_1(D)$. The slice Cauchy–Schwarz inequality already present in Plan 2 gives, in this centered annular cap,

$$|D \Delta B_1|^2 \leq C \, M_1(D) \tag{4.3}$$

with a computable absolute constant when $\eta \leq \eta_0$. Thus a very natural sufficient estimate for (AB) is

$$E(D) - E(B_1) \geq c \, M_1(D). \tag{4.4}$$

For radial graphs, (4.4) is exactly what the direct graph proof establishes after removing the neutral volume and translation modes. For general annular sets, (4.4) is not yet proved in the repo. The missing analytic lemma is a capacitary radial-moment coercivity estimate:

$$\int_D |\nabla z|^2 \geq \left( \frac{1}{4} + c \right) M_2(D) \quad \text{modulo the volume and translation neutral modes.} \tag{4.5}$$

The hard part is not high frequency by itself, nor translation. Translation is the first harmonic and must be quotiented out or fixed by centering. High-frequency radial corrugations are already handled by the slope-free radial-graph theorem. The hard part is multi-crossing geometry in the centered annulus:

- the net signed radial displacement on each ray should be controlled by the same Steklov spectral gap as in the graph proof;
- same-ray cancellations, where $D$ has both an inner missing piece and an outer excess piece on the same direction, should cost local radial capacity, plausibly linearly in their cancelled volume;
- no file in Plan 2 or Plan 3 currently proves this capacitary decomposition for the torsion energy.

Plan 2's slice-rearrangement material supplies the same-center geometric technology behind (4.3): per-ray good/bad decompositions, bad-ray multiplicity absorption, oriented radial trace, and estimates of the form $|E\Delta B(z)|^2 \le C\Pi(E,z)$ under an annular cap. It does not supply a single-domain torsion lower bound converting $E(D) - E(B)$ into one of those same-center annular defects. That is the precise remaining bridge.

# 6 Soundness verdict

- **Overclaim found.** Chain B's unweighted Hessian inequality $(T)$ is false. The false claim is in the transition
$$\delta_T(D) \quad \Longrightarrow \quad \int_D |D^2 u_D + \tfrac{1}{n} I|^2,$$
not in Step 1, Step 3, or Step 5.
- **Underclaim corrected.** Lack of $C^1/H^1$/curvature control is not by itself fatal in dimension 2. The new radial-graph theorem shows that small annular amplitude can be enough if one can first prove single-crossing, or replace single-crossing by an annular-set theorem.
- **Centering caveat.** Any annular theorem must remove the translation mode. The fixed-center bound by $|D\Delta B_1|^2$ is false without barycenter/optimal-center normalization, as translated disks show.
- **Best non-Plan-1 targets.** Prove the planar annular entry (AE), then prove the planar annular bridge (AB), or the stronger radial-moment coercivity (4.4). This would use the correct $H^{1/2}$/Fraenkel scale and would not rely on BDV.
- **What not to pursue.** Do not try to close Plan 3 through a single-level Hessian square function, weighted or unweighted. Its spectral order is too high.